

Mathematical Modeling Assignment 1

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1 Cost optimisation for a consultancy firm

1.1 Basic Problem

A consultancy company obtained five new contracts with customers located in Lund, Karlskrona, Linköping, Umeå and Karlstad. To deliver on these contracts, the company needs to hire new staff. The company has offices in Malmö, Gothenburg and Stockholm. To minimise the cost, at which office should the new staff be hired?

The five contracts require a different number of working hours per year. The following working hours per year have been agreed in the contracts:

Customer location	Hours required
Lund	500
Karlskrona	1200
Linköping	2400
Umeå	3900
Karlstad	800

Table 1: Annual hours required by contract.

It is estimated that a staff member works 1500 hours per year. Each contract is for one year. The hiring costs per new staff member vary, depending on the location of the office. The following amounts in SEK have been estimated:

	Malmö	Gothenburg	Stockholm
Hiring cost per new staff member (SEK)	1 300 000	1 800 000	2 400 000

Table 2: Hiring costs per staff member.

It is more efficient to work with customers which are located in close proximity. The following table gives an estimate how much it costs to deliver a staff year (i.e. 1500 staff hours) to the different customers from each office. Numbers are in SEK per staff year.

A staff member can work on multiple projects during the year, for any fraction of their time. Only full-time positions can be offered; no part-time positions are allowed.

Customer	Malmö	Gothenburg	Stockholm
Lund	38 000	524 000	1 204 000
Karlskrona	418 000	676 000	992 000
Linköping	834 000	550 000	396 000
Umeå	2 490 000	1 964 000	1 270 000
Karlstad	1 066 000	498 000	616 000

Table 3: Cost to deliver one staff year = 1500 hours.

1.2 How many staff members should each office hire?

How many staff members should be hired at each office to minimise the total costs? Use the five-step method. Model as a mixed integer linear programming model. Some variables of the problem are continuous, while others are integer. Please state the total costs. Is each office using all the staff time it gains from the new hires?

According to the five-step method, we first identify the problem by identifying the variables:

Symbol	Description	Type	Units
Decision variables			
h_i	Number of new staff-years hired in office i ($i \in \{m, g, s\}$ for Malmö, Gothenburg, Stockholm)	Integer	staff-years
x_{ij}	Total hours dispatched from office i to customer j , $j \in \{\text{Lund}, \dots, \text{Karlstad}\}$	Continuous	hours
Parameters			
D_j	Hours demanded by customer j	Parameter	hours
C_i^{hire}	Cost to hire one staff-year in office i	Parameter	SEK/staff-year
C_{ij}^{disp}	Cost to dispatch one staff-year from office i to customer j	Parameter	SEK/staff-year
Auxiliary variables			
S_i	Total hours dispatched from office i	Continuous	hours
T_j	Total hours received by customer j	Continuous	hours
H_i	Dispatch capacity of office i	Continuous	hours
C	Total cost	Continuous	SEK

Table 4: All variables and parameters

We next highlight the relationships between the variables from table above:

Table 5: Relations and Constraints Between Variables

Relations and Constraints

$$\begin{aligned}
S_i &= \sum_j x_{i,j} \\
T_j &= \sum_i x_{i,j} \\
T_j &\geq D_j \\
S_i &\leq 1500 h_i \\
h_i &\in \mathbb{Z}_{\geq 0} \\
x_{i,j} &\geq 0 \\
C &= \sum_i C_i^{\text{hire}} h_i + \sum_{i,j} C_{ij}^{\text{disp}} x_{i,j}
\end{aligned}$$

Our objective of this problem is then to minimize the total cost C . The next step is to choose the correct modeling approach. In our case, we have both the total cost C and constraints as linear objectives. Notice that h_i must be integer (whole staff-years), x_{ij} can be fractional (hours can be split arbitrarily). Therefore we can choose either MILP solver in scipy or Simplex + Integer solver in excel(used here). The next step is to enter the information inside excel as in Figure 1:

	A	B	C	D	E	F	G	H	I	J	K
1											
2											
3			Cost to hire new staff in different locations. Per year, per 1500 hours								
4											
5											
6											
7			Malmö	Gothenburg	Stockholm						
8		C ^{hire} _i	1300000	1800000	2400000						
9			h _m	h _g	h _s			Useable hours from each office (H _i)			
10		Hired h _i staff. (Discrete)	3	1	2			4500	1500	3000	
11											
12			Cost to let staff hired dispatch to target city. Per year, per 1500 hours								
13											
14											
15											
16		C ^{disp} _{ij}	Malmö	Gothenburg	Stockholm						
17		Lund	38000	524000	1204000						
18		Karlskrona	418000	676000	992000						
19		Linköping	834000	550000	396000						
20		Umeå	2490000	1964000	1270000						
21		Karlstad	1066000	498000	616000						
22											
23											
24											
25			Solution								
26											
27		x _{ij}	Malmö	Gothenburg	Stockholm	Sum T _j		Hours needed (D)			
28		Lund	500	0	0	500		500			
29		Karlskrona	1200	0	0	1200		1200			
30		Linköping	2400	0	0	2400		2400			
31		Umeå	200	700	3000	3900		3900			
32		Karlstad	0	800	0	800		800			
33		Sum S _i	4300	1500	3000						
34											
35		Total Cost(C)	16235600								
36											

Figure 1: Excel sheets of the problem.

By applying the Simplex + Integer solver in excel we get a minimized total cost of 16235600 SEK by hiring 3 new staffs from Malmö, 1 new staff from Gothenburg and 2 new staffs from Stockholm. Observe that this optimized solution has used all staff hours

available from Gothenburg and Stockholm offices with a 200 hours slack from Malmö office.

1.3 Shadow prices

There are a number of constraints associated with the problem. Specify for which constraint there is and for which constraint there isn't a shadow price. Give an explanation. Calculate the shadow prices.

Note that we have solved the question in both excel and scipy, in this section we calculate the shadow prices using scipy. This is done by recalculate the total cost, but add 1 to each constraint each time. The difference in cost will then be the shadow price of the respective constraint. The results are shown in Figure 2 and Figure 3:

```
new cost: 16235625.333333334
shadow price for Lund : 25.333333333954215

new cost: 16235878.666666668
shadow price for Karlskrona : 278.66666666790843

new cost: 16236155.999999998
shadow price for Linköping : 555.9999999981374

new cost: 16237260.0
shadow price for Umeå : 1660.0

new cost: 16236282.666666666
shadow price for Karlstad : 682.6666666660458
```

Figure 2: Shadow price of customer constraint.

```
new cost: 16235599.999999994
shadow price for Malmö : -5.587935447692871e-09

new cost: 16235249.333333332
shadow price for Gothenburg : -350.66666666790843

new cost: 16233973.333333333
shadow price for Stockholm : -1626.666666669771
```

Figure 3: Shadow price of working hour constraint in the 3 offices.

As can be seen, only the constraint of Malmö working hours has no shadow price. This is not surprising, as from Figure 1, there are 200 slack hours from Malmo. Hence, increasing the maximum working hours of each employee in Malmo would not yield additional costs, as they would not need to use the extra hour. By the same token, because there are no slack hours in Gothenburg and Stockholm, they would have shadow prices. On the other hand, there are shadow prices for all the required hours of every customer because employees from Malmo will have to work 1 extra hour, incurring an additional cost by Table 3.

The customer in Umeå requires more work. What happens when the customer in Umeå requires a total of 4100 hours, 4300 hours or 4500 hours? Use the shadow prices to calculate how this would change the cost estimate. What cost estimate do you obtain when using the solver to estimate these three cases? Why do the results not always agree?

We calculate the costs for 4100, 4300, 4500 hours using the shadow price of Umeå, and by using the solver. Both results are shown in Figure 4 and Figure 5, respectively.

```

calculating new cost via shadow price

# Calculated umea shadow price earlier
shadow_price = 1660
earlier_required_hours = 3900
earlier_total_cost = 16235600.0

new_hours = [4100, 4300, 4500]
for i in range(len(new_hours)):
    new_cost = (new_hours[i] - earlier_required_hours) * shadow_price + earlier_total_cost
    print("New cost for", new_hours[i], ":", new_cost)

New cost for 4100 : 16567600.0
New cost for 4300 : 16899600.0
New cost for 4500 : 17231600.0

```

Figure 4: Cost calculated via shadow price

```

Total cost: 16567600.0
Number of employees hired in each office: [3. 1. 2.]
Total cost: 18199600.0
Number of employees hired in each office: [4. 1. 2.]
Total cost: 18460000.0
Number of employees hired in each office: [ 4. -0.  3.]

```

Figure 5: Cost calculated via solver

The shadow price in the case of 4100 hours is the same for both methods as the nature of the solution does not change. Recall our base-case solution ($h_m = 3, h_g = 1, h_s = 2$). As can be seen in Figure 5, the nature of the solution is the same as in ($h_m = 3, h_g = 1, h_s = 2$).

However, in the case of 4300 and 4500 hours for Umeå, the nature of the solution changes to ($h_m = 4, h_g = 1, h_s = 2$) and ($h_m = 4, h_g = 0, h_s = 3$) respectively. Hence, the cost estimate for both methods differ, with the cost estimate calculated via shadow price

no longer being accurate, and the estimate via solver reflecting the true cost estimate instead.

1.4 Gothenburg candidates

The searches at the Gothenburg office have produced two outstanding candidates. How would hiring at least two candidates in Gothenburg change the cost estimate. Use the required hours as originally stated.

Again we model the information and other constraints this in excel as in section 1.2, since we need to hire at least two candidates in Gothenburg we simply just add one more constrain $H_g \geq 2$ and apply the Simplex + Integer solver again. The results are shown in Figure 6:

		Cost to hire new staff in different locations. Per year, per 1500 hours								
		Malmö	Gothenburg	Stockholm						
	C^hire _ i	1300000	1800000	2400000						
	Hired h_i staff. (Discrete)	h_m	h_g	h_s						
		3	2	1						
		Cost to let staff hired dispatch to target city. Per year, per 1500 hours								
	C^disp _ ij	Malmö	Gothenburg	Stockholm						
	Lund	38000	524000	1204000						
	Karlskrona	418000	676000	992000						
	Linköpping	834000	550000	396000						
	Umeå	2490000	1964000	1270000						
	Karlstad	1066000	498000	616000						
		Solution								
	x_ij	Malmö	Gothenburg	Stockholm	Sum T_j	Hours needed (D)				
	Lund	500	0	0	500	500				
	Karlskrona	1200	0	0	1200	1200				
	Linköpping	2400	0	0	2400	2400				
	Umeå	200	2200	1500	3900	3900				
	Karlstad	0	800	0	800	800				
	Sum S_i	4300	3000	1500						
	Total Cost(C)	16329600								

Figure 6: Excel modeling with at least 2 hired staff in Gothenburg.

Recall our base-case solution ($h_m = 3, h_g = 1, h_s = 2$) with total cost 16235600, the new hires vector became (h_m, h_g, h_s) = (3, 2, 1)

The recalculated total minimized cost is 16329600 SEK with a cost difference: $16329600 - 16235600 = 94000$. Thus, forcing two hired staffs in Gothenburg raises the total annual cost by 94000 SEK.